LAB 1 — REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #11 — Healthcare & Telemedicine

Telemedicine Slot Booking & Prescription Portal

1. Functional Requirements

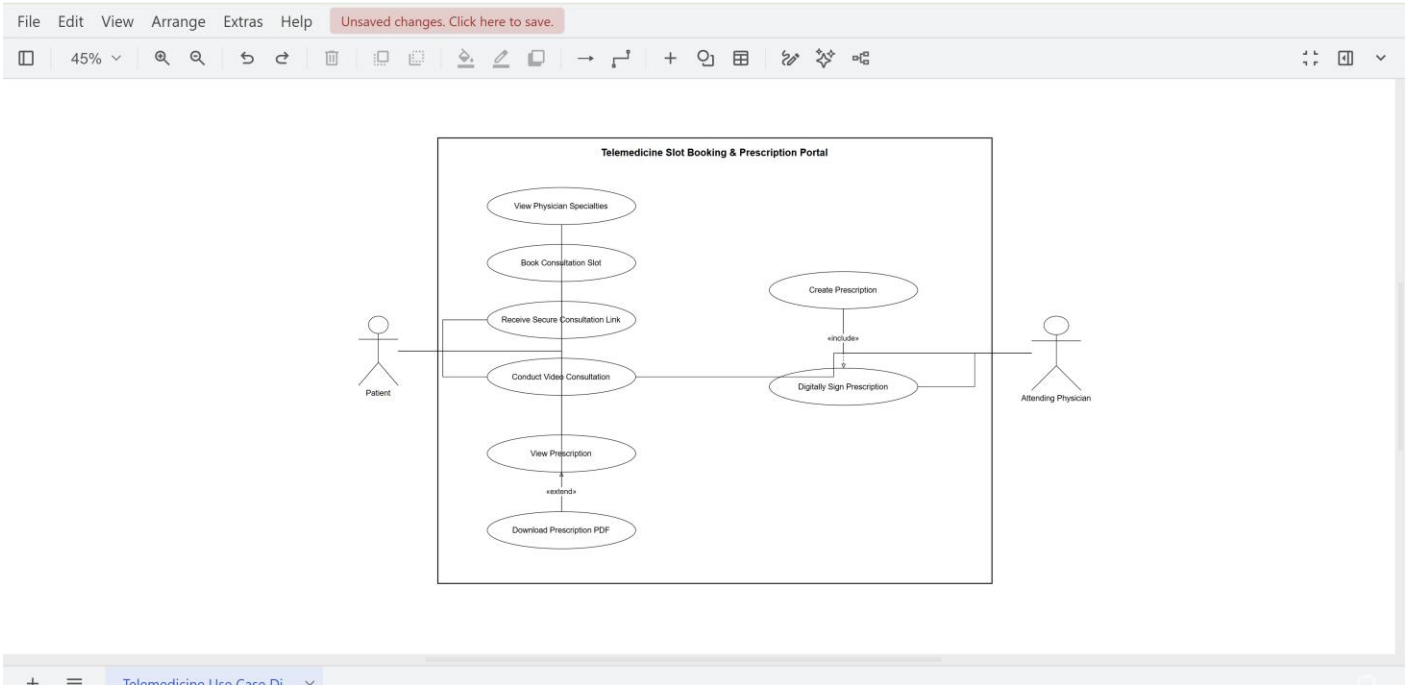
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable patients to view attending physicians and their medical specialties.	High	Pass: Patient can view physician names and specialties. Fail: Physician specialty information is unavailable or incorrect.	Helps patients select an appropriate physician before booking.
FR-002	Functional	The system shall enable patients to view available consultation slots and book a video consultation with an attending physician.	High	Pass: A selected available slot is successfully booked and associated with the patient. Fail: The system allows booking of an unavailable slot.	Provides the core telemedicine appointment functionality.
FR-003	Functional	The system shall provide patients with a secure consultation-room link after successful slot booking.	High	Pass: A valid secure consultation-room link is generated and provided to the patient. Fail: No link is generated or the link is accessible without authorization.	Enables patients and physicians to conduct the scheduled remote consultation securely.
FR-004	Functional	The system shall enable attending physicians to create and digitally sign structured prescriptions for patients after consultation.	High	Pass: Prescription contains the physician's valid digital signature. Fail: Prescription is generated without physician license validation.	Ensures prescriptions are authored and authenticated by the attending physician.
FR-005	Functional	The system shall enable patients to download their digitally signed prescriptions as PDF records.	High	Pass: Patient can download a readable PDF containing the digitally signed prescription. Fail: PDF cannot be downloaded or lacks the required signature.	Provides patients with a usable digital record of their prescription.

2. Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	All video-session metadata and patient clinical notes shall be encrypted in transit using TLS 1.3 and at rest using AES-256.	High	Pass: Security testing confirms TLS 1.3 is used for data in transit and AES-256 for stored data. Fail: Any protected data is transmitted or stored without the specified encryption.	Protects sensitive healthcare information from unauthorized access.
NFR-002	Performance	The system shall maintain acceptable response latency and security performance under simulated peak load.	High	Pass: Benchmarking tests confirm the defined target latency and security standards under simulated peak load. Fail: The system fails to meet the target latency or security standards.	Ensures the portal remains usable and secure when many users access it simultaneously.

3. UML USE-CASE DIAGRAM

Black-and-white UML diagram for the Telemedicine Slot Booking & Prescription Portal.



4. USE-CASE FLOW SPECIFICATION

Core Use Case: Book Consultation Slot

Preconditions

The patient has access to the telemedicine portal.
Attending physician specialties and available consultation slots are available in the system.
The patient is eligible to book a consultation.

Postconditions

A selected consultation slot is successfully reserved for the patient.
The booking is associated with the attending physician and patient.
A secure consultation-room link is made available to the patient after successful booking.

Main Success Scenario

1. Patient opens the telemedicine portal.
2. Patient views attending physicians and their medical specialties.
3. Patient selects an appropriate attending physician.
4. System displays the physician's available consultation slots.
5. Patient selects an available slot.
6. Patient confirms the booking.
7. System records the consultation booking.
8. System generates a secure consultation-room link.
9. System provides the consultation-room link to the patient.

Alternate Flow

At Step 5, if the selected consultation slot is no longer available, the system informs the patient that the slot cannot be booked and displays the currently available slots. The patient selects another available slot and continues from Step 6.